



U.S. Department of Transportation
Federal Highway Administration

Turner-Fairbank
Highway Research Center

Intelligent Transportation Systems (ITS) Analysis, Modeling and Simulation (AMS) Community of Practice

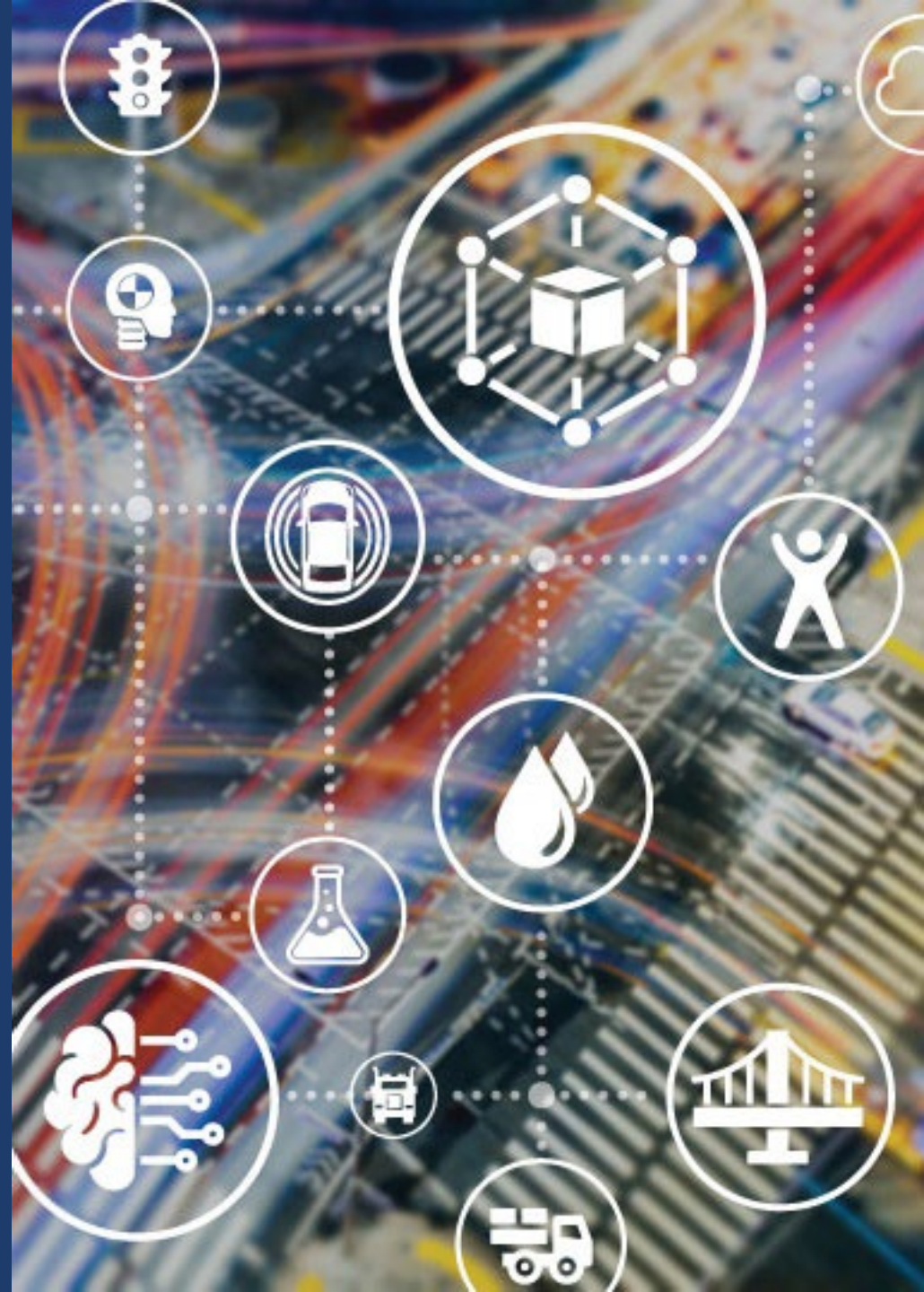
John Hourdos and Jongsun Won: Federal Highway Administration (FHWA)

Hyungjun Park: ITS Joint Program Office (JPO)

ITS AMS Community Initial Member Meeting Day 1

December 2, 2025

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Welcome and Introduction

John Hourdos, FHWA



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Introductions

- ▶ Federal team introduces themselves.
- ▶ Attendees enter the following information in meeting chat:
 - ▷ Name.
 - ▷ Title.
 - ▷ Entity.



Initial Member Meeting Menti Instructions⁽¹⁾ (1/2)

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Who Is Here Today

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What is your involvement in AMS?



- Academia
- State or local agency
- Consultant
- USDOT
- Vendor
- Non-USDOT Federal partner.
- Other

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What is your involvement in AMS?



Purpose of Today's Event

- ▶ To introduce the purpose of this community.
- ▶ To reach agreement on the community charter:
 - ▷ The community's mission, goals, and objectives.
 - ▷ How the community will conduct business.
- ▶ To identify initial ITS AMS community subgroups and task forces.



Detailed Day 1 Agenda

Time (Eastern)	Topics	Speakers and Facilitators
01:00–01:15 p.m.	Welcome and Introduction	John Hourdos, FHWA
01:15–01:45 p.m.	How Did We Get Here?	John Hourdos, FHWA
01:45–02:00 p.m.	AMS-Related Programs	Hyungjun Park, John Hourdos, and Jongsun Won, FHWA
02:00–02:45 p.m.	Guest Speaker**	Karl Wunderlich, Noblis
02:45–03:00 p.m.	Break	
03:00–03:30 p.m.	Walk Through the Initial ITS AMS Community Collaboration Framework and Solicit Feedback**	Kathy Thompson, Noblis
03:30–04:00 p.m.	Identify Community Task Forces**	John Hourdos, FHWA
04:00 p.m.	Adjourn	John Hourdos, FHWA

**Designates member participation.



Detailed Day 2 Agenda

Time (Eastern)	Topics	Speakers and Facilitators
01:00–01:15 p.m.	Recap of Day 1	John Hourdos, FHWA
01:15–02:15 p.m.	Panel Discussion	Sanhita Lahiri, Virginia Department of Transportation Daniel Morgan, Caliper Corporation Samer H. Hamdar, The George Washington University
02:15–02:25 p.m.	Break	
02:25–03:15 p.m.	Task Force Group Activity**	John Hourdos, FHWA
03:15–03:45 p.m.	Introduction to the Community Forum**	Rick Grahn, Noblis
03:45–03:55 p.m.	Revisit Parking Lot, Additional Feedback, and Questions**	John Hourdos, FHWA
03:55–04:00 p.m.	Next Steps and Upcoming Community Meetings	John Hourdos, FHWA
04:00 p.m.	Adjourn	John Hourdos, FHWA

**Designates member participation.



FHWA's Vision

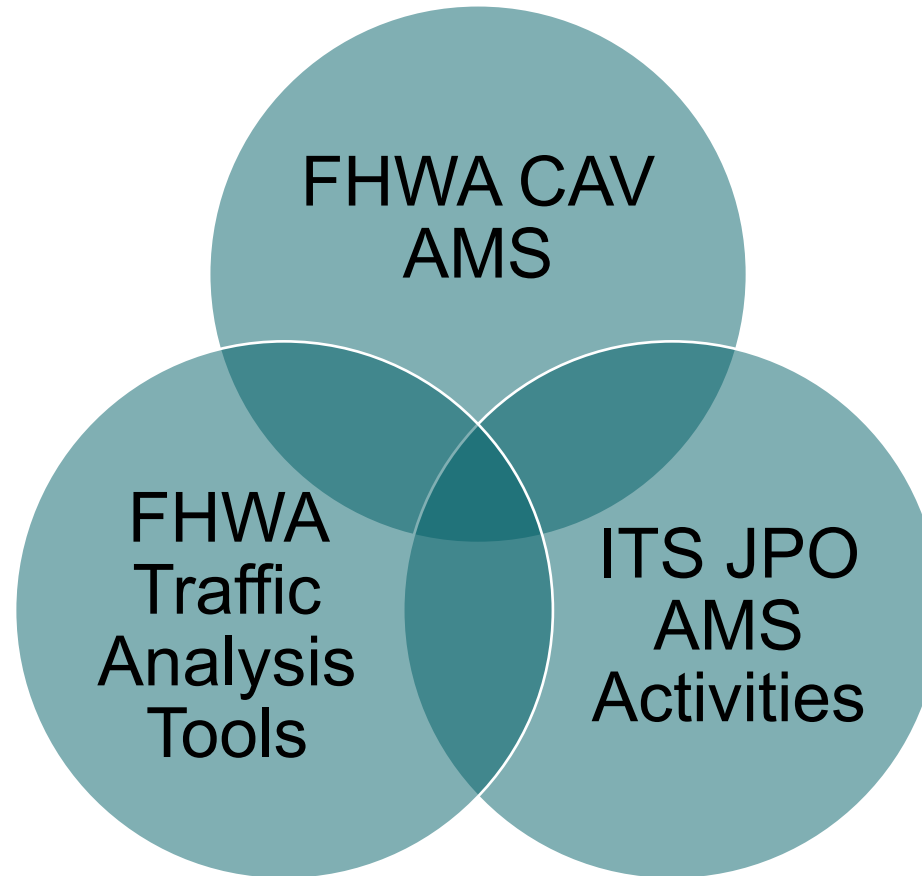
Improve AMS of ITS, including connected and automated vehicles (CAVs).



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Cross-Functional Coordination



Source: FHWA.



Purpose of This Community

- ▶ Bring together professionals and stakeholders engaged in ITS AMS activities.
- ▶ Encourage knowledge exchange on tools, methods, and applications.
- ▶ Identify shared challenges and opportunities for innovation.
- ▶ Promote coordination among agencies, researchers, and practitioners.





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How Did We Get Here?

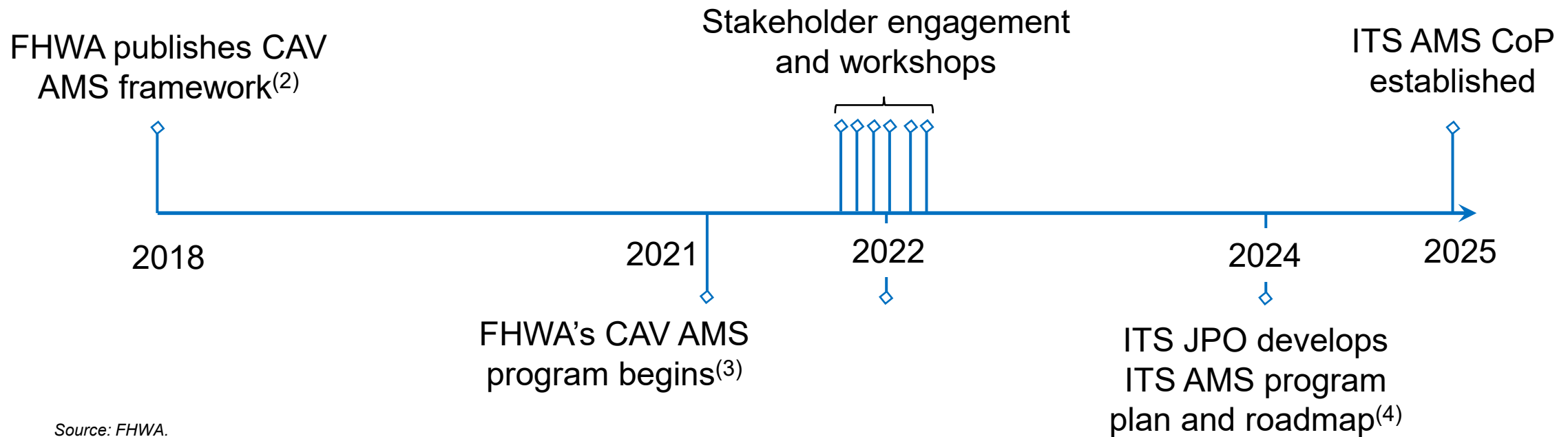
John Hourdos, FHWA



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History and Motivation for Community of Practice (CoP)



Source: FHWA.



Stakeholder Engagement Overview

Event or Activity	Date	Target Stakeholder Group	Number of Participants*
Stakeholder Event 1	2/04/2022	State and local practitioners who use AMS tools on a regular basis.	23
Stakeholder Event 2	2/25/2022	AMS tool vendors and consultants who develop methodologies.	19
Stakeholder Event 3	2/28/2022	Researchers who may have research solutions to incorporate CAV into AMS.	14
Stakeholder Event 4	4/08/2022	The three groups listed in the first three rows and those interested in program direction.	104
Federal Workshop	4/26/2022	Identified Federal staff.	29
FHWA Workshop	5/16/2022	Identified FHWA staff.	16

*The number of participants includes the FHWA participants but does not include the hosts or the program support staff.



Needs Identified by Stakeholders (1/4)

Develop Guidance

Modeling, Cost/Benefit, Scenario Planning, Safety, Calibration, MOEs, Assumptions

Conduct Behavioral Research

Driver, Automated Vehicle, Work Zone Safety, Fleet Penetration

Establish and Update Standards

Shared Data Format, CAV Definitions, Cybersecurity

Collect and Disseminate Data

Trajectory, Storage, Standardization, Lane Change, Car Following

Characterize System Impacts

Penetration, Mixed, Capacity, Demand, Extreme Conditions, Infrastructure

Map Research Needs to Potential Tools and Workflows

Open Behavior Model, Tool Library

Engage Stakeholders

Industry, Academia, State/Local, Education, Collaboration, Expert Panels, Facilitation

Research and Develop Use Cases and Pilot Projects

Emerging Tech, Vulnerable Road User Impacts

Research Human Factors

Adoption Rates, Decisionmaking

Source: FHWA.

MOE: Measures Of Effectiveness.



Needs Identified by Stakeholders (2/4)

Engage Stakeholders

Industry, Academia, State/Local, Education,
Collaboration, Expert Panels, Facilitation

Source: FHWA.



Needs Identified by Stakeholders (3/4)

Develop Guidance

Modeling, Cost/Benefit, Scenario Planning,
Safety, Calibration, MOEs, Assumptions

Engage Stakeholders

Industry, Academia, State/Local, Education,
Collaboration, Expert Panels, Facilitation

Source: FHWA.



Needs Identified by Stakeholders (4/4)

Develop Guidance

Modeling, Cost/Benefit, Scenario Planning,
Safety, Calibration, MOEs, Assumptions

Establish and Update Standards

Shared Data Format, CAV Definitions,
Cybersecurity

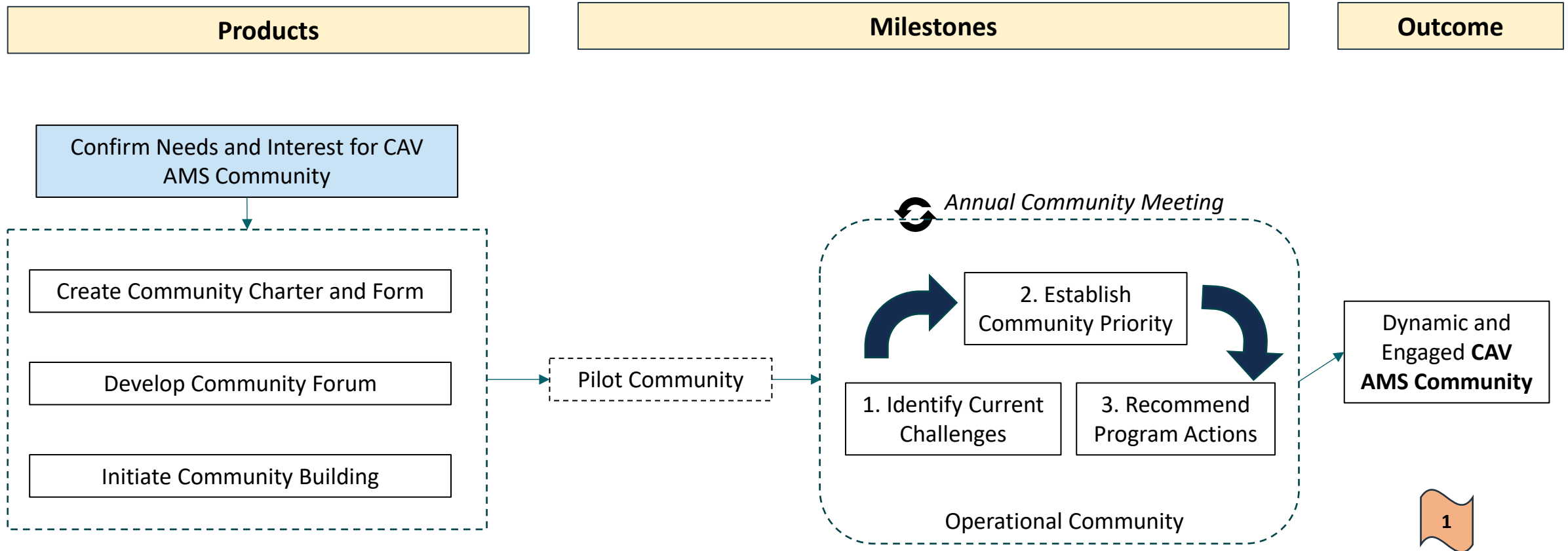
Engage Stakeholders

Industry, Academia, State/Local, Education,
Collaboration, Expert Panels, Facilitation

Source: FHWA.



Establishing a CoP



Source: FHWA.





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AMS-Related Programs

Hyungjun Park, ITS JPO

John Hourdos, FHWA

Jongsun Won, FHWA



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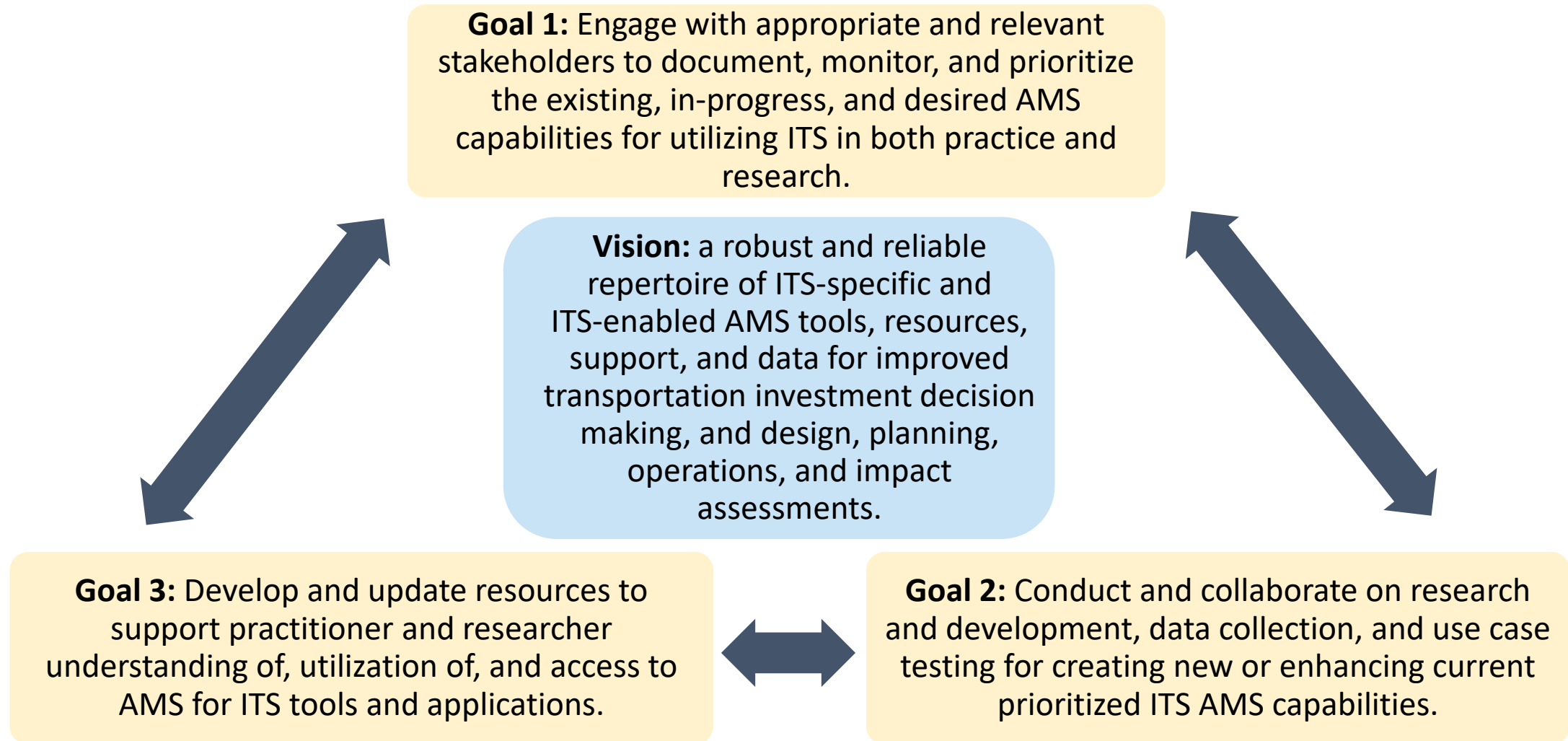
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Scope of ITS JPO's AMS for ITS Program

- ▶ Methodologies, algorithms, use cases, datasets, data exchanges, tools, guidance, policy, pilot deployments, evaluations, and professional capacity building activities that incorporate at least one ITS component (see the national ITS Reference Architecture for Cooperative and Intelligent Transportation (ARC-IT) Version 9.2).⁽⁵⁾
- ▶ Broad spectrum of ITS that encompasses a wide range of technologies at different stages of research and development (R&D) and readiness for deployment:
 - ▷ **Highly developed, well-understood ITS:** Including traveler and transit information systems, dynamic real-time routing, electronic toll collection, ramp metering, and traffic signal management systems. Successful adoptions of these ITS have been achieved by numerous U.S. transportation agencies with documented benefits, costs, and lessons learned.
 - ▷ **Less developed, less understood ITS:** Including areas such as connected vehicles (CVs), automated vehicles (AVs), cooperative driving automation, and the corresponding communications, as well as the incorporation of advanced analysis techniques (e.g., machine learning) for operational strategy improvements and cybersecurity protection. Additional R&D is needed before these ITS components are considered ready for mainstream adoption and integration.



Strategic Direction



Source: FHWA.



Research Tracks

- ▶ **Research Track 1**—Foundational research and development: Includes activities that are exploratory and new in nature regarding AMS for ITS topics and stem from the community engagement and program oversight efforts.
- ▶ **Research Track 2**—Data collection, integration, and management: Includes data-specific activities that stem from the community engagement, program oversight efforts, and research activities.
- ▶ **Research Track 3**—Model development and enhancement, case studies, and testing: Encompasses activities that build on fundamental research, often incorporating data-related activities that aim to meet the needs and address the gaps and challenges identified from community engagement and program oversight.
- ▶ **Research Track 4**—Outreach, resources, and knowledge and technology transfer: Includes activities intended to convey information regarding the findings from activities' various audience types.



Scope of FHWA CAV AMS Program Plan

► Background:

- ▷ CAV technologies have the potential to significantly improve system safety and efficiency.
- ▷ CAV technologies are expected to impact both strategic traveler behavior and tactical driving behaviors.
- ▷ AMS tools must be available to explore CAV impacts and their interactions at the system level.

► Key gaps:

- ▷ Lack of proven, practical modeling methodologies.
- ▷ Data-related gaps.
- ▷ Difficulty integrating system-level changes into CAV AMS tools.
- ▷ Lack of tools and guidance.



Program Vision and Goals

► Vision:

Support State and local agencies, the predominant infrastructure owners and operators in the United States, in understanding the most likely impacts of CAVs on roadways through key performance metrics (e.g., safety and mobility) to support improved operations and investment decisionmaking.

► Goals:

- ▷ Coordinate CAV AMS efforts across FHWA's disciplines (e.g., operations, planning, safety, policy, freight, and infrastructure).
- ▷ Advance understanding of the impacts of CAVs on roadways according to key performance metrics (e.g., safety and mobility).
- ▷ Accelerate development of AMS tools and models (ranging from traffic microscopic to macroscopic models), including improved functionality and validation.



Program Plan's Desired Outcomes

ID	Title	Description
1	Active CAV AMS Community	Establish a dynamic and engaged CAV AMS community connecting key CAV stakeholders, AMS stakeholders, and FHWA to surface key issues, focus resources and attention, and participate in CAV AMS program activity.
2	Track CAV Technology Capabilities and Distribution	Track the functions and capabilities of CAV vehicle and infrastructure technology, including the distribution (both fleet and infrastructure penetration) of these technologies across the Nation's surface transportation system.
3	Human Behavior Response to Different Types of CAV Technology	Conduct and synthesize research regarding human traveler and driver behavior observed or reported by individuals reacting to different types of CAV technologies in different environments.
4	Validated System Data for CAV AMS Collected and Shared	Share validated system data for CAV modeling in a standard format to support CAV AMS tool development efforts.
5	Regularly Update CAV AMS Tool Resources	CAV AMS Tool Resources provide possible future mixed fleet and technology scenarios and range of uncertainties, how these scenarios and ranges can be incorporated into model development, and potential benefits and impacts of CAV technologies.

ID = identifier.



Program Tracks

- ▶ CAV AMS community engagement.
- ▶ Technology tracking and human behavior research.
- ▶ Data collection and sharing.
- ▶ CAV AMS methodologies development and case studies.
- ▶ CAV AMS tool resources.



Vision

Support stakeholders in the traffic analysis community in the United States by providing necessary guidance and support to achieve effective and efficient traffic analysis and advance the traffic analysis community to achieve confident and reliable traffic analysis outcomes.



Goals

1. Provide guidance and support for the efficient and effective use of traffic analysis tools for decision support.
2. Develop new or improve existing traffic analysis tools and supporting materials to incorporate evolving stakeholders' needs.
3. Broaden and strengthen the traffic analysis tools community.



Traffic Analysis Tools Program

Vision



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A well-supported traffic analysis community with access to needed resources to sufficiently understand and perform traffic analysis tasks and provide ***confident and reliable traffic analysis results*** to the decisionmakers will produce the following benefits:

Goals



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- ▶ Provide support for the efficient and effective use of traffic analysis tools.
- ▶ Develop new and improve existing traffic analysis tools to incorporate evolving expectations and needs.
- ▶ Broaden and strengthen the traffic analysis community.



Priorities



Keep toolbox products up to date and catalog toolbox products



Make program awareness efforts—Spread the word



Strengthen the traffic analysis community



Balance focus areas (tool categories and travel modes)

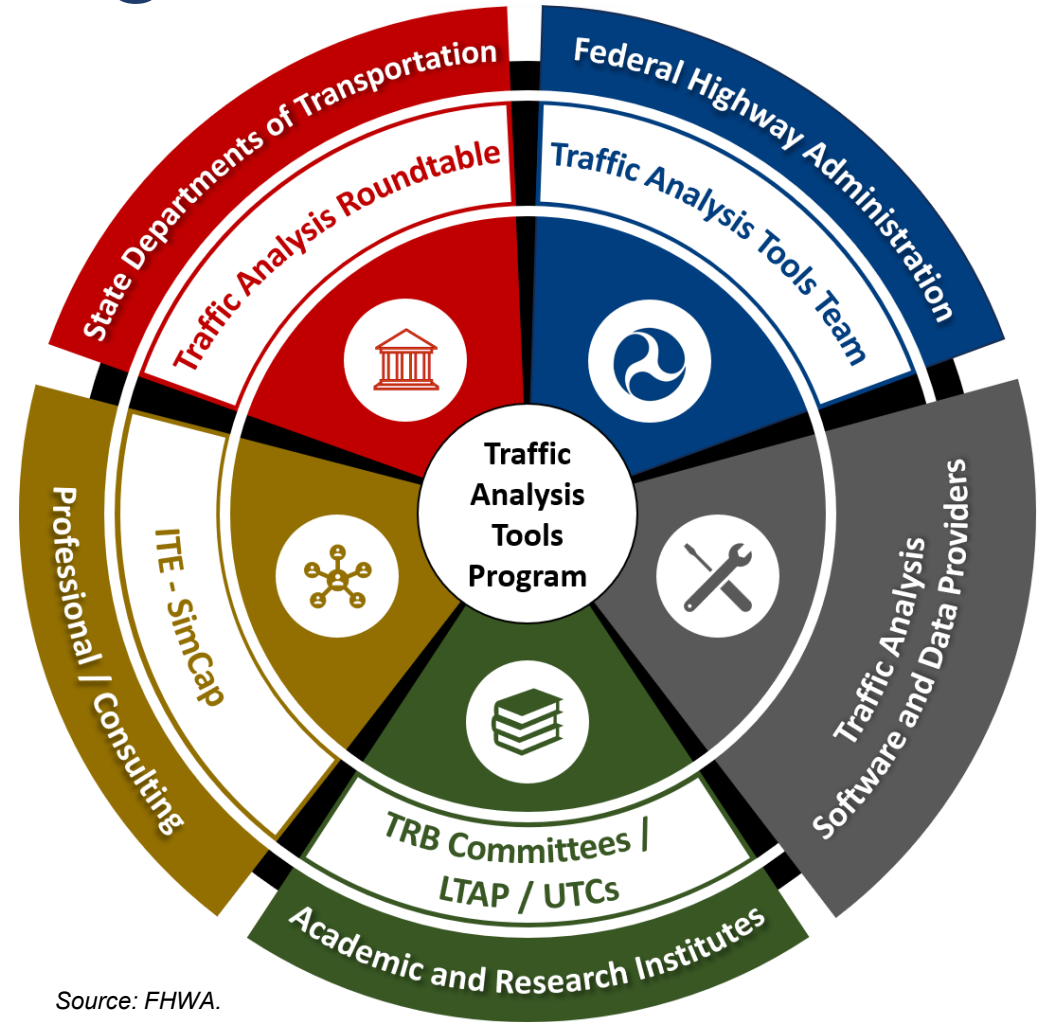


Return to basics (users, tools, data, etc.)

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Traffic Analysis Tools Program Stakeholders



Source: FHWA.

ITE = Institute of Transportation Engineers; SimCap = Simulation and Capacity Analysis User Group; TRB = Transportation Research Board; LTAP = Local Technical Assistance Program; UTC = U.S. Department of Transportation's University Transportation Centers Program.





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Guest Speaker

Karl Wunderlich, Noblis



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Advancing ITS AMS Through Community Action: A 21st Century Perspective

Karl Wunderlich, Noblis





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Break



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Walk Through the ITS AMS Community Charter and Solicit Feedback

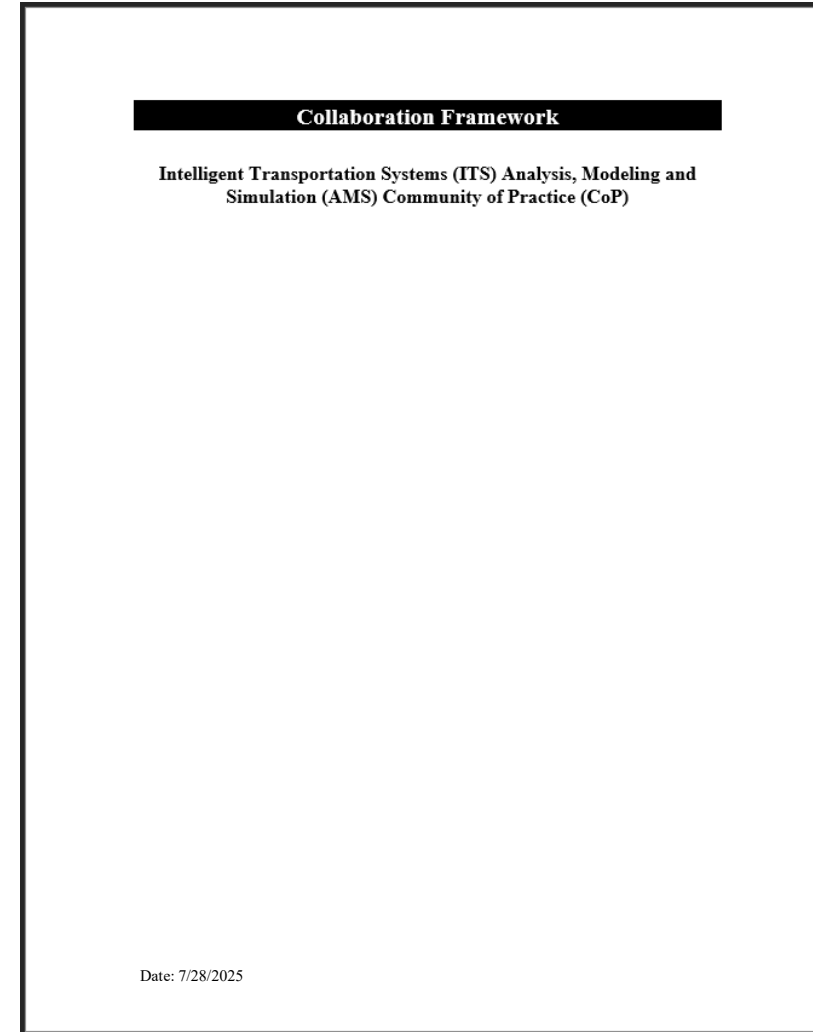
Kathy Thompson, Noblis



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Collaboration Framework



Source: USDOT.





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Identify Community Task Forces

John Hourdos, FHWA



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Potential Task Force Topics

Planning

Operations

Technology (CV, AV, and
infrastructure)

Cross-cutting (data,
comms, cyber, etc.)



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Adjourn



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Contacts

John Hourdos

John.Hourdos@dot.gov

(202) 493-3491

Gene McHale

Gene.McHale@dot.gov

(202) 493-3275

Jongsun Won

Jongsun.Won@dot.gov

(202) 366-1327

Hyungjun Park

Hyungjun.Park@dot.gov

(202) 366-9126



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